

SIMATS ENGINEERING

Name	Register Number	Course Code	Course
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13. Perform Perspective Transformation on the Video.

AIM:

Perspective Transformation on the Video.

PROGRAM:

```
import cv2
import numpy as np

roi_points = np.array([
    (150, 200),
    (450, 200),
    (550, 500),
    (50, 500)
], dtype=np.float32)

target_points = np.array([
    (0, 0),
    (400, 0),
    (400, 600),
    (0, 600)
], dtype=np.float32)

M = cv2.getPerspectiveTransform(roi_points, target_points)

cap = cv2.VideoCapture(0)

if not cap.isOpened():
    print("Error: Could not open webcam.")
    exit()

while True:
    ret, frame = cap.read()

    if not ret:
        print("Error: Could not read frame.")
        break
```

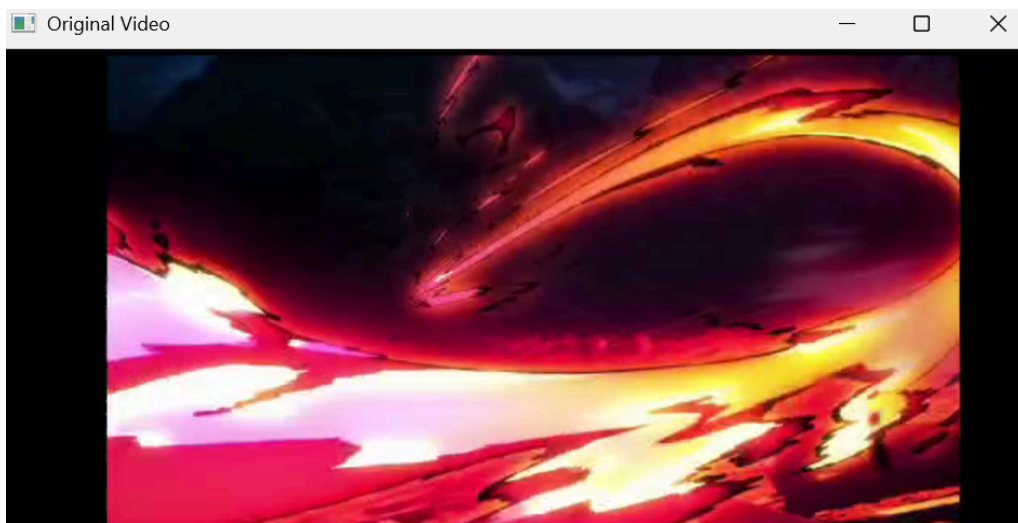
```
dst = cv2.warpPerspective(frame, M, (400, 600))

cv2.imshow("Original Frame", frame)
cv2.imshow("Transformed Frame", dst)

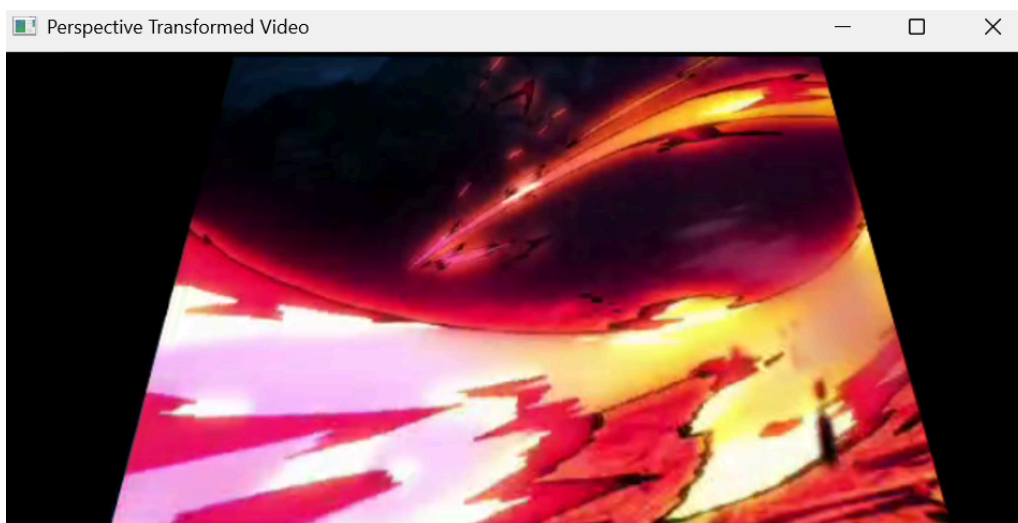
if cv2.waitKey(1) & 0xFF == ord('q'):
    break

cap.release()
cv2.destroyAllWindows()
```

INPUT:



OUTPUT:



RESULT:

Thus the program is executed successfully.